Firmware Uploading to WiseNode

Before starting, please go through [LoRaWAN End Node documentation]() or skip it if you already have firmware binary or hex file.

Software you need:

- STM32CubeProgrammer

Hardware you need:

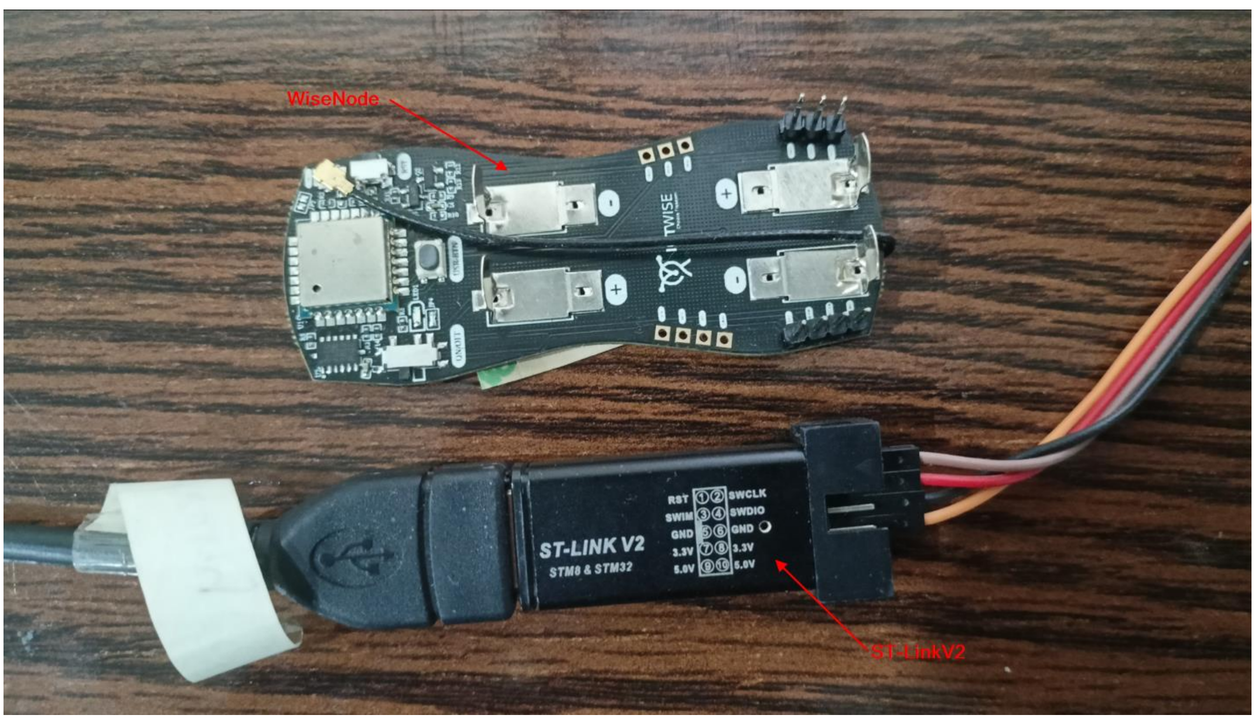
- ST-LinkV2
- WiseNode

Install the above listed software if you don't have them already installed.

Following are steps to upload the firmware to WiseNode:

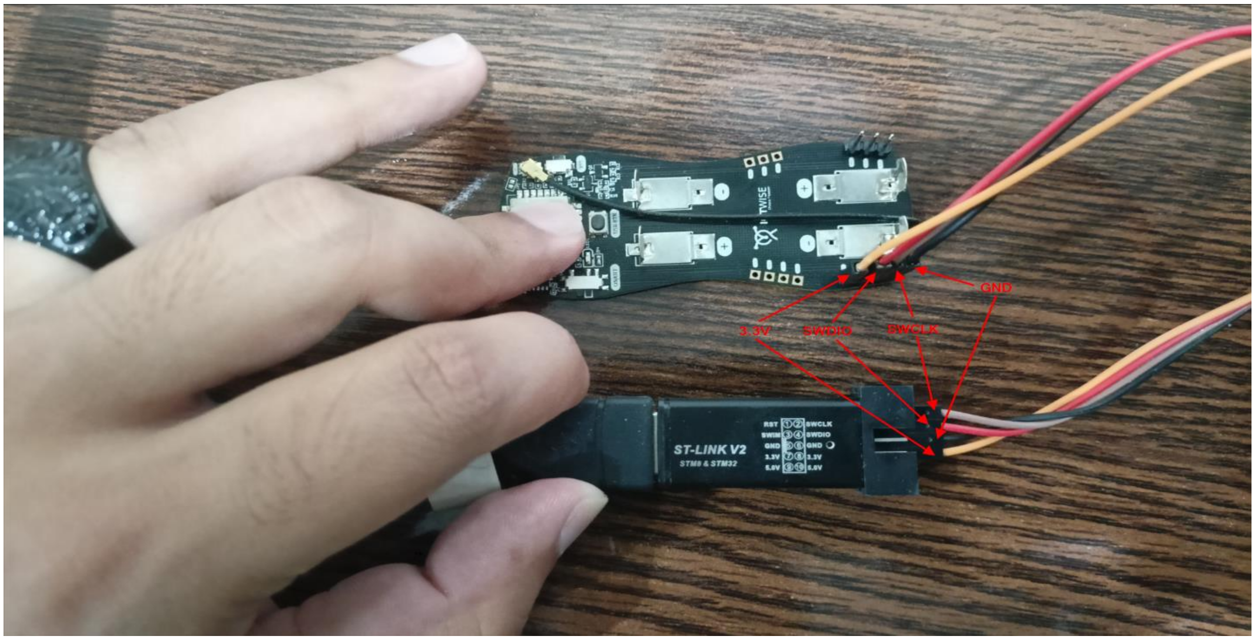
Step 1:

First of all, prepare a ST-LinkV2 and WiseNode hardware which you want to program. Solder the male headers onto the WiseNode as they can be seen in the figure below:



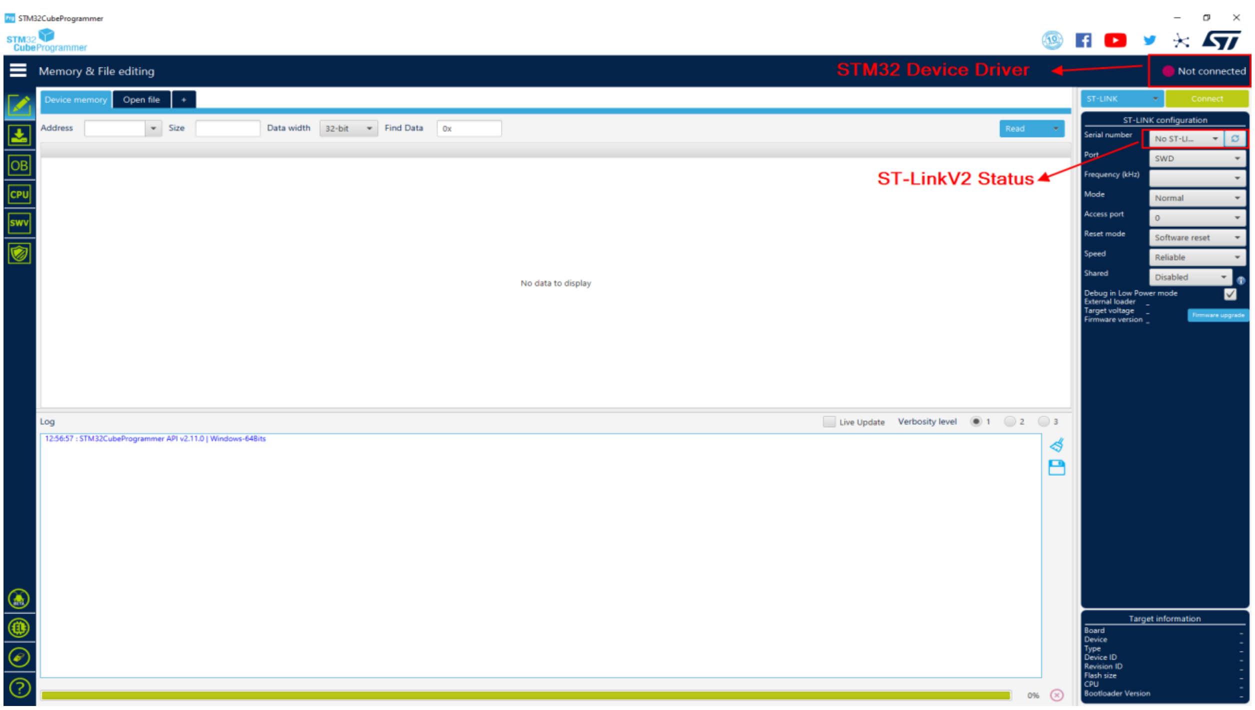
Step 2:

Make the connections between ST-LinkV2 and WiseNode as shown in the figure below:



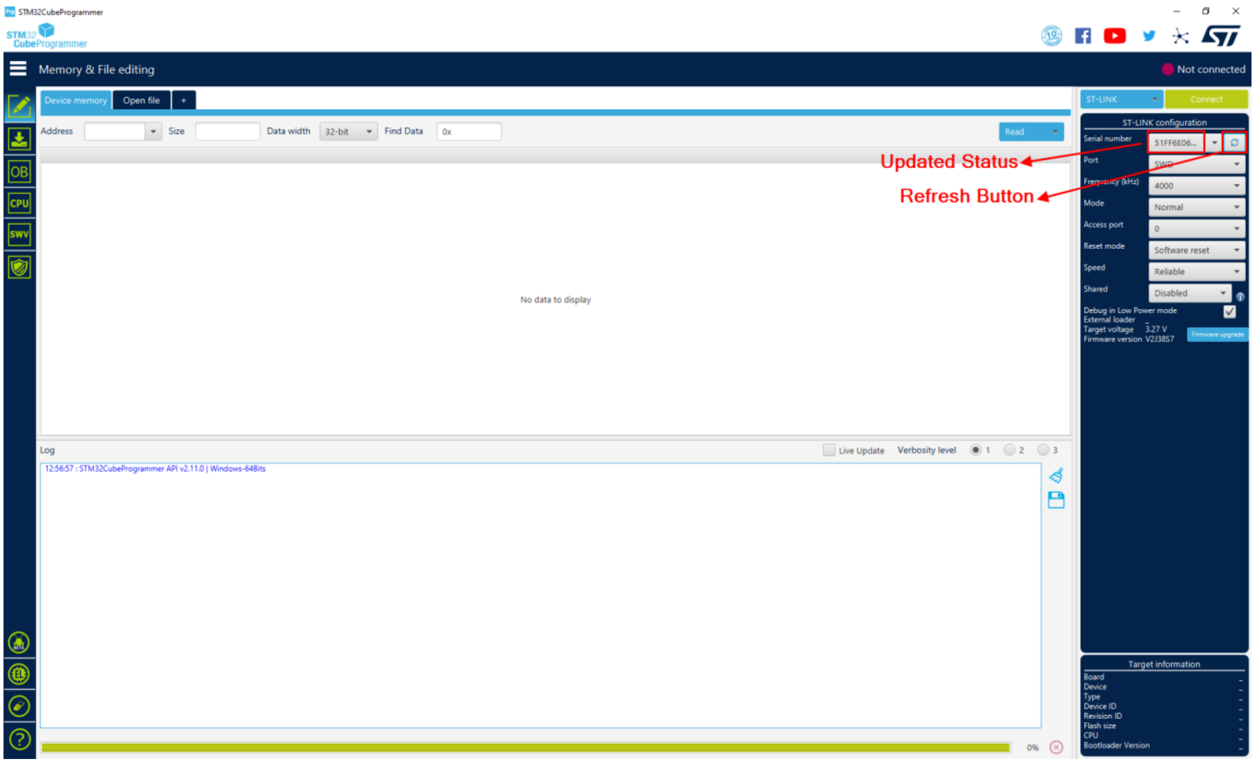
Step 3:

Open the STM32CubeProgrammer.



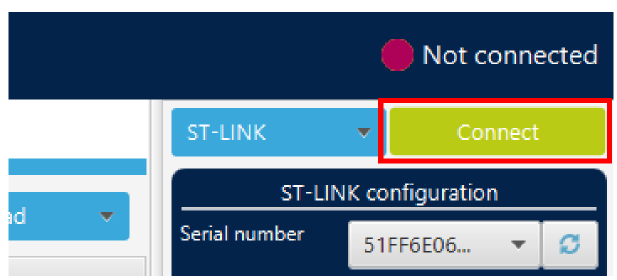
Step 4:

Now connect the ST-LinkV2 with your computer. Make sure you have ST-LinkV2 drivers installed on your computer. In STM32CubeProgrammer, click “Refresh” icon to update the status of ST-LinkV2.



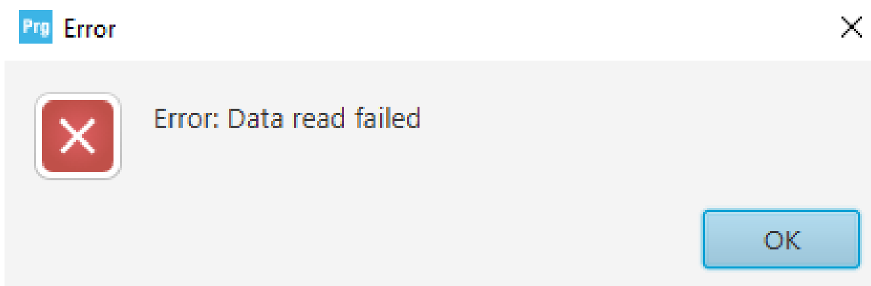
Step 5:

Now press the “Connect” button to connect with STM32 device (WiseNode in our case).



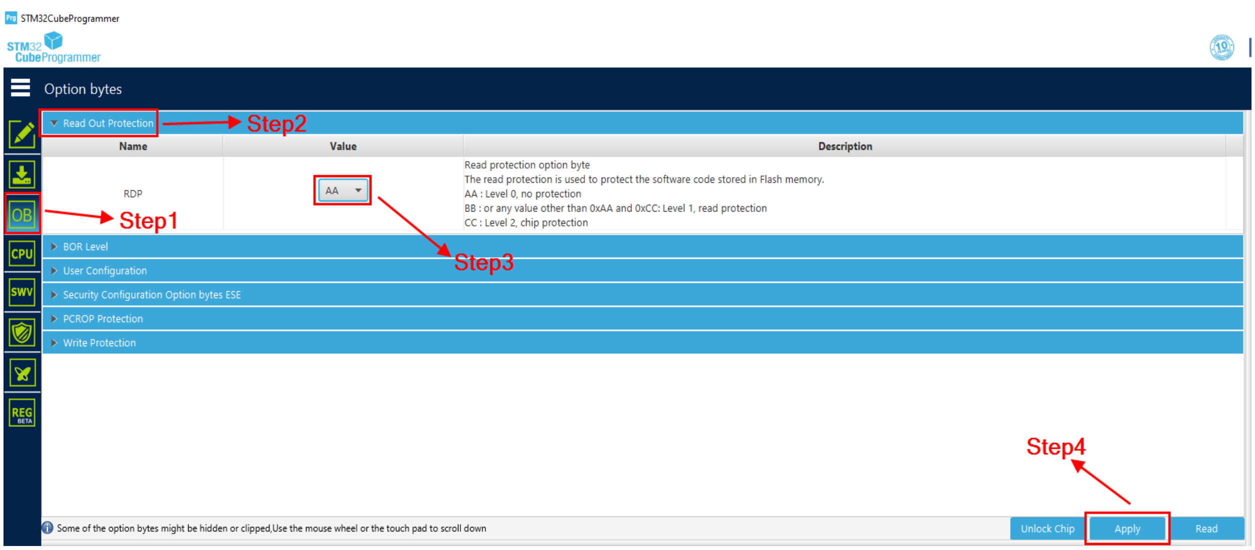
If the device gets connected, the status will be updated.

You’ll get this error if you are connecting a new device for the very first time. New devices come with read protection. So, to communicate with it, first we have to unlock the device.

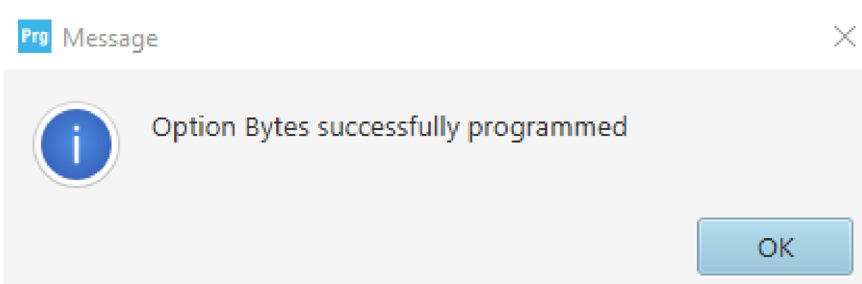


Step 6:

To unlock the device, follow the steps as shown in the figure below.



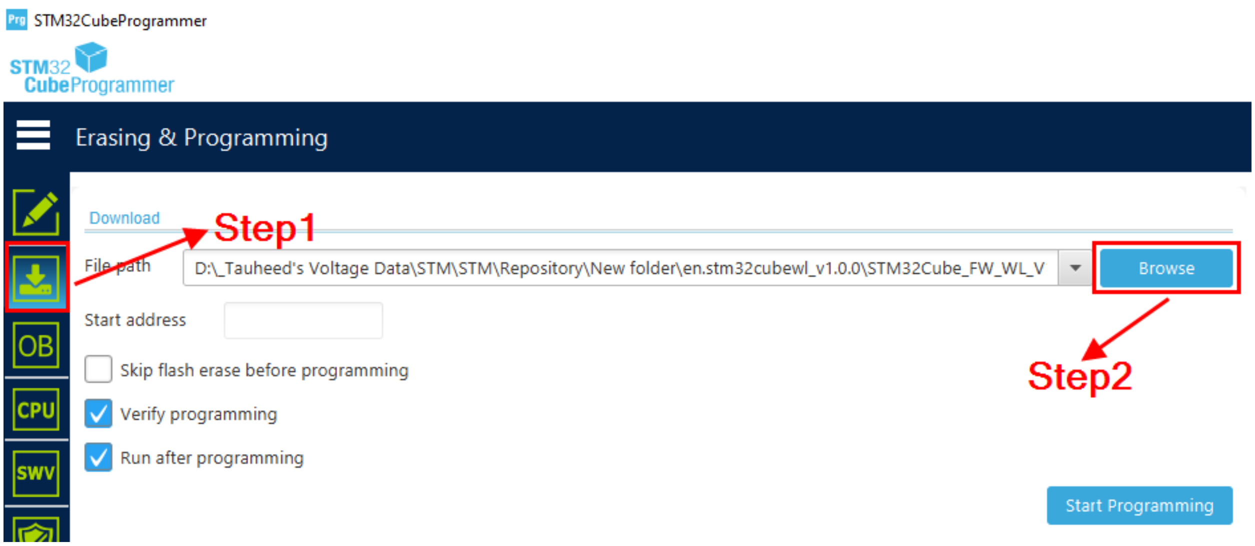
You’ll get the confirmation once it’s done.



Now, we are ready to upload the firmware file.

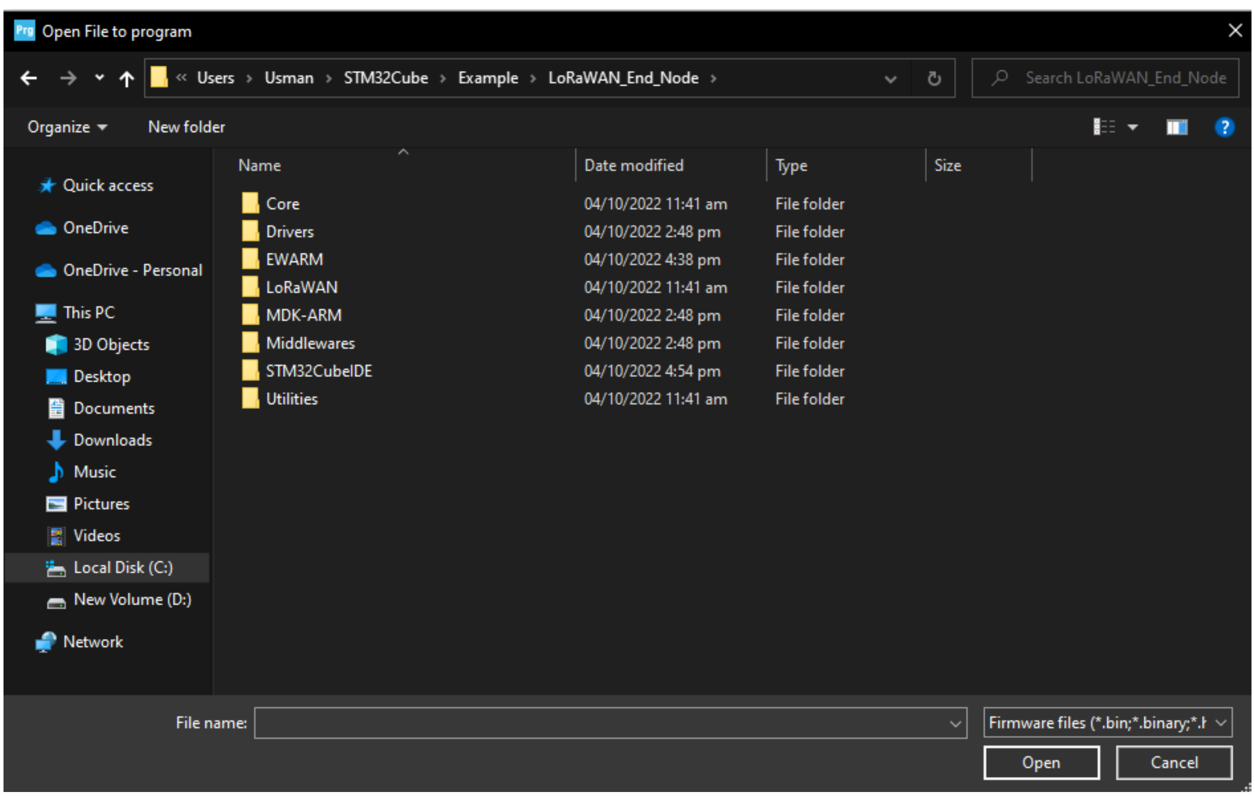
Step 7:

Go to the Erasing and Programming section and click “Browse” button as shown in figure.



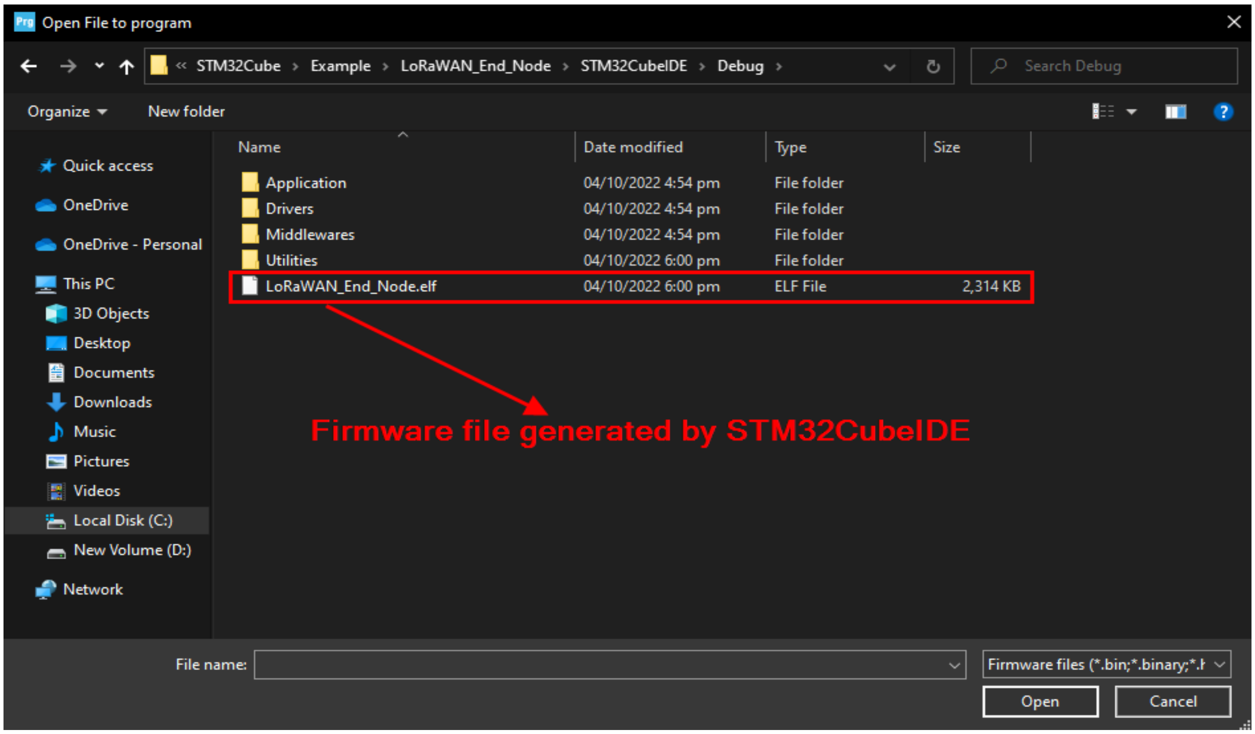
Step 8:

Navigate to the LoRaWAN_End_Node project directory.



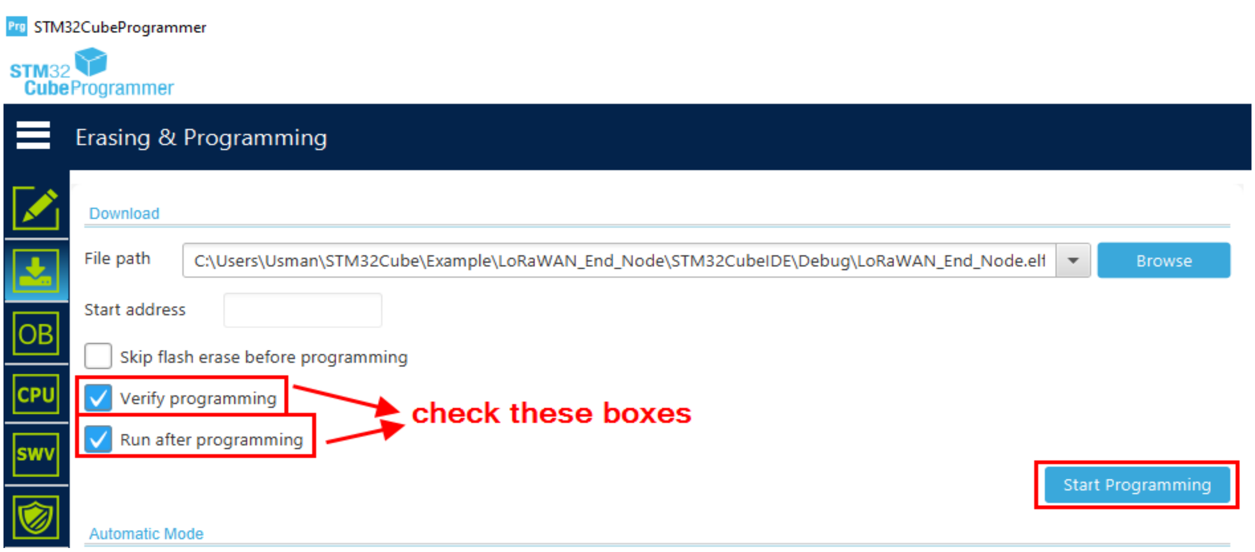
Step 9:

Now open the directory "STM32CubeIDE>Debug" and select the "LoRaWAN_End_Node.elf" file and open it.



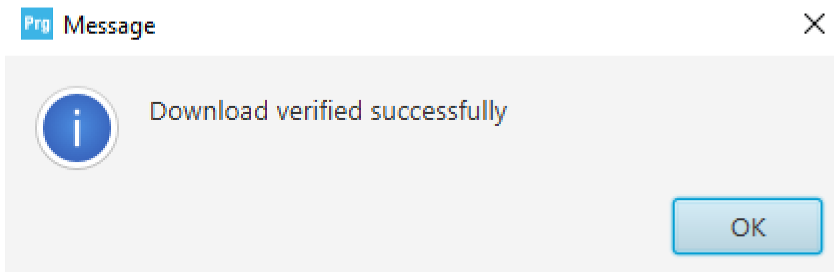
Step 10:

Make sure to check the "Verify programming" and "Run after programming" boxes and then click the "Start Programming" button.



Step 11:

After programming, you'll get the confirmation message in a dialog box.



Enjoy now as the WiseNode is ready to communicate on a Button Interrupt with Helium network via gateway.